

1.1.1. Calculate Momentum 06:15

Write a program that accepts the mass of an object (in kilograms) and its velocity (in meters per second), then calculates and displays the momentum of the object. The momentum p is calculated using the formula:

$$p = m \times v$$

where:

- m is the mass of the object (in kilograms).
- v is the velocity of the object (in meters per second).

Input Format:

- A single floating-point number representing the mass of the object in kilograms.
- A single floating-point number representing the velocity of the object in meters per second.

Output Format:

- The output will display calculated momentum with appropriate units (kgm/s) (rounded up to 2 decimal places).

Sample Test Cases +

calculate... Submit

Explorer

1 mass = float(input())
2 velocity = float(input())
3 momentum = mass * velocity
4 print(f"momentum: .2fkgm/s")

Debugger

Terminal

Test cases

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1.1.2. Conditional Calculation Based on the Number of Digits

21:20

Write a Python program that accepts an integer n as input. Depending on the number of digits in n .

Constraints:
 $1 \leq n \leq 999$

Input Format:

- The input consists of a single integer n .

Output Format:

- If n is a single-digit number, print its square.
- If n is a two-digit number, print its square root (rounded to two decimal places).
- If n is a three-digit number, print its cube root (rounded to two decimal places).
- Else print "Invalid".

Sample Test Cases

condition...

```
1 import math
2 n = int(input())
3 if 1 <= n <= 999:
4     if 1 <= n <= 9:
5         print(n * n)
6     elif 10 <= n <= 99:
7         print(f"math.sqrt(n):.2f")
8     else:
9         print(f"{n ** (1/3):.2f}")
10 else:
11     print("Invalid")
```

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1.1.3. Days between Two Dates05:33

Write a Python program to calculate number of days between two dates.

Input Format:

- The first line contains the first date in the format `YYYY-MM-DD`.
- The second line contains the second date in the format `YYYY-MM-DD`.

Output Format:

- An integer representing the number of days between the given dates.

Note:

- The first date should always be considered earlier than the second date.

Sample Test Cases

noOfDays...

```
1 from datetime import datetime
2 date1 = input().strip()
3 date2 = input().strip()
4 date1 = datetime.strptime(date1, "%Y-%m-%d")
5 date2 = datetime.strptime(date2, "%Y-%m-%d")
6 difference = (date2 - date1).days
7 print(difference)
```

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1.1.4. Reverse a Number

24/02

Write a Python program to reverse the digits of the number and print the reversed number.

Input Format:

The input is a single integer.

Output Format:

The output prints a single integer, which is the reversed number.

Sample Test Cases

reverseN...

1

n = int(input())

2

rev = 0

3

while n > 0:

4

digit = n % 10

5

rev = rev * 10 + digit

6

n = n // 10

7

print(rev)

Terminal

Test cases

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1.1.5. Multiplication Table02:46

Write a Python program that takes an integer as input and prints the multiplication table for that integer from 1 to 10.

Input Format:

- The first line of input contains an integer that represents the number for which the multiplication table is to be printed.

Output Format:

- Print the multiplication table for the given number in the format:

<number> x <multiplier> = <result>

Note:

- The character "x" is used in the output to represent multiplication.
- Refer to the visible test-cases for more clarity.

Sample Test Cases

multiplica...

1n = int(input())

2for i in range(1,11):

3 print(f"{n} x {i} = {n*i}")

TerminalTest cases

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1.2.1. Pass or Fail

19:22

Write a Python program that accepts the number of courses and the marks of a student in those courses.
The grade is determined based on the aggregate percentage:

- If the aggregate percentage is greater than 75, the grade is Distinction.
- If the aggregate percentage is greater than or equal to 60 but less than 75, the grade is First Division.
- If the aggregate percentage is greater than or equal to 50 but less than 60, the grade is Second Division.
- If the aggregate percentage is greater than or equal to 40 but less than 50, the grade is Third Division.

Input Format:

- The first input will be an integer n , the number of courses.
- The second input will be n integers representing the marks of the student in each of the n courses, separated by a space.

Output Format:
If the student passes all courses:

- Print the aggregate percentage (formatted to two decimal places).
- Print the grade based on the aggregate percentage.

If the student fails any course (marks < 40 in any course), print:

- "Fail".

Note:

- Aggregate Percentage: Average performance across all courses, calculated by summing all marks and dividing by the number of courses.

passorFa...

Submit

```
1 def calculate_grade(num_courses, marks):
2     if len(marks) != num_courses:
3         return "Error: Number of marks does not match number of courses."
4     if any(mark < 40 for mark in marks):
5         return "Fail"
6     aggregate_percentage = sum(marks)/num_courses
7     if aggregate_percentage > 75:
8         grade = "Distinction"
9     elif aggregate_percentage > 60:
10        grade = "First Division"
11    elif aggregate_percentage > 50:
12        grade = "Second Division"
13    elif aggregate_percentage >= 40:
14        grade = "Third Division"
15    else:
16        grade = "Fail"
17    return f"Aggregate Percentage: {aggregate_percentage:.2f}\nGrade: {grade}"
18    try:
19        num_courses = int(input().strip())
20        marks = list(map(int, input().strip().split()))
21        result = calculate_grade(num_courses, marks)
22        print(result)
23    except ValueError:
24        print("Error: Invalid input. Please enter integers only.")
```

TerminalTest cases

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1.2.2. Fibonacci Series 03:48 AA 🌙 📄 🔗 —

Write a Python program that uses recursion to print the first n terms of the Fibonacci series.

Input Format:

- A single integer n representing the number of terms to generate.

Output Format:

- A single line containing the first n terms of the Fibonacci sequence, separated by spaces.

Sample Test Cases

+

fibonacci.... 🔍 Submit

```

1  def fibonacci(n):
2      if n == 1:
3          return 0
4      elif n == 2:
5          return 1
6      else:
7          return fibonacci(n - 1) + fibonacci(n - 2)
8  n = int(input())
9  for i in range(1, n + 1):
10     print(fibonacci(i), end=" ")
11

```

Terminal Test cases

1.2.3. Pattern - 1 02:19

Write a Python program to print a pattern of asterisks in the form of a right-angled triangle.

Input Format:

- The input is an integer, representing the number of rows in the pattern.

Output Format

- The output should display the pattern of asterisks (*), with each row containing an increasing number of asterisks.

Note: Refer to the displayed test cases for the sample pattern.

Sample Test Cases +

rightangl...

Submit

```

1 n = int(input())
2 for i in range(1, n + 1):
3     for j in range(i):
4         print("*", end=" ")
5     print()

```

Terminal Test cases

Debugger

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1.2.4. Pattern - 2 10:27

Write a Python program to print a right-angled triangle pattern of numbers.

Input Format:

- The input is an integer, representing the number of rows in the pattern.

Output Format:

- The output should display the pattern of numbers separated by space, with each row containing increasing numbers starting from 1 up to the row number.

Note:

- Refer to the displayed test cases for the sample pattern.

Sample Test Cases +

numberP... Submit

```

1 n = int(input())
2 for i in range(1,n+1):
3     for j in range(1,i+1):
4         print(j, end=" ")
5     print()
    
```

Terminal Test cases